ABOUSSAAD OTHMANE

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I am a Master of Research student in Modelling, Optimization & Decision (PSL Research University), in parallel with a Science Master Degree in Applied Mathematics & Computer Science for Data Science (CY Tech - CY Cergy Paris University). My research interests lie at the intersection between Optimization, Dynamical Systems and Manifold Learning for Uncertainty Quantification.

Education

PSL Research University (#1 in France, #6 in Europe and#24 Worldwide - QS 2025)

Master in Modeling, Optimization, Decision and Organization (MODO) *Sept. 2025 - Aug. 2026*

Master of Research focusing on mathematical modelling, optimization and decision sciences, supported by **Université Paris-Dauphine - PSL**'s Laboratory for Analysing and Modelling Systems for Aiding Decisions (LAMSADE, UMR CNRS7243) and **Mines Paris - PSL**'s Center for Management Science (CGS, UMR CNRS9217), undertaken in parallel with my final year at CY Tech. Accepted with an Excellence Grant from **École Normale Supérieure - PSL (ENS Ulm)**.

Courses: Graphs and applications, Mathematical Programming, Modelling in Decision Aiding - Operational Research, Preference Modelling and Multicriteria Decision Making, Industrial and Design Models, Decision under Uncertainty, Multiobjective Optimization, Problem Structuring Methods, Stochastic Programming, Algorithms for Continuous Optimization.

CY Tech - CY Cergy Paris University (#5 Computer Science School in France - Le Figaro 2026)

Applied Mathematics and Computer Science Engineering Program *Sept. 2023 - Aug. 2026*

Relevant Courses: Probability, Statistics, Algorithmics, Databases, Data Mining, Optimization, Numerical Analysis, Graph Theory, AI, Time Series, PDE.

Result: Top 10% of the class

CPGE IBN GHAZI, Rabat, Morocco

Preparatory Classes for Grandes Écoles (Maths-Physics) *Sept. 2020 - June 2023*

A rigorous program preparing students for entrance exams to France's elite engineering schools, focusing on advanced mathematics and physics.

Relevant courses:

- **Math:** Algebra, Linear Algebra, Calculus, Probability, Differential Calculus
- **Physics:** Quantum Physics, Mechanics, Signal Processing, Optics, Electromagnetism, Thermodynamics
  Result: Top 10% of the class

Experience and Projects

Research Intern, Imperial College London, UK *April 2025 - August 2025*

- Visiting Student at the Department of Mathematics under the supervision of Dr. Boumediene Hamzi (**Caltech, Imperial College London and The Alan Turing Institute**). Worked on **kernel methods** for high-dimension diffusion operators and **statistical learning**.
- Developed approaches to **learn data-adaptive kernels** with analysis of **spectral stability** and **scalability** (approximation techniques). *(Manuscript in preparation)*

Research Project, CY Cergy Paris University, France *October 2024 - April 2025*

- Academic research project at the **ETIS Lab (Information Processing and Systems Team, UMR CNRS 8085)** under the supervision of Dr. Sonia Yassa (**CY Cergy Paris University**).
- Worked on implementing a **surrogate multi-objective optimization model** with **probabilistic models** for costly evaluations in IoT (Edge/Fog/cloud settings). *(Manuscript in preparation)*

Full-stack Developer Intern, CDG Group, Rabat, Morocco *July 2024 - August 2024*

The **Moroccan Deposit and Management Fund (CDG Group)** is Morocco's sovereign investment and public asset management institution. With numerous subsidiaries across various economic sectors, CDG required a comprehensive solution to streamline its operations. As part of this initiative, I developed a shared services e-catalogue solution for Groupe CDG:

- Developed a shared services e-catalogue using Angular, Spring, JWT, and MySQL, serving over **60 subsidiaries**.
- Integrated demo request functionality with secure authentication via JWT.
- Managed and tested back-end services using Postman and Scrum methodology.

Research and Publications

Publication: Learning Kernels in Diffusion Maps (Internship at Imperial College London) *Finalizing*

- **Co-authors: Othmane Aboussaad**, Adam Miraoui, Houman Owhadi, Boumediene Hamzi.
- **Affiliations:** Department of Mathematics, Imperial College London, London, UK; The Alan Turing Institute, London, UK; Department of Computing and Mathematical Sciences, Caltech, CA, USA.

Publication: Multi-objective Optimization in Industrial Workflows *Finalizing*

- **Co-authors: Othmane Aboussaad**, Sonia Yassa.
- **Affiliations:** ETIS Lab, UMR CNRS 8051, CY Cergy Paris Université, Cergy, France.

Publication: A Survey of Human-Centered Artificial Intelligence: Explainable Anomaly Detection for Cybersecurity in Smart Manufacturing 5.0 *Finalizing*

- Literature Review on ethical AI applications and cybersecurity for human-centered technologies at the **AGM Laboratory (Geometry, Analysis and Modelling, UMR8088)**.
- **Co-authors: Othmane Aboussaad**, Thi Hien Nguyen.
- **Affiliations:** Laboratoire AGM, UMR CNRS 8088, CY Cergy Paris Université, Cergy, France.

Publication: Effect of the measurement errors on one-Sided Synthetic-RZ control charts for monitoring the ratio of two normal variables *Finalizing*

- Study aiming to assess the impact of measurement errors on the performance of one-sided Synthetic-RZ control charts used for monitoring the ratio of two normal variables at the **AGM Laboratory**.
- **Co-authors: Othmane Aboussaad**, Thi Hien Nguyen, Kim Duc Tran, Kim Phuc Tran.
- **Affiliations:** Laboratoire AGM, UMR CNRS 8088, CY Cergy Paris Université, Cergy, France; Institute of Artificial Intelligence and Data Science, Dong A University, Danang, Vietnam; École Nationale Supérieure des Arts et Industries Textiles, GEMTEX Laboratory, Roubaix, France

Certifications, Awards and Distinctions

Cambridge Centre for AI in Medicine – ML Summer School 2025 *September 2025*
Summer School organized by Prof. Mihaela van der Schaar (**University of Cambridge**) on AI/ML for healthcare: an intensive 5-day program covering LLMs, generative models, causality, data-centric AI, time series, uncertainty, digital twins, AutoML, and responsible AI, with clinical applications.

Hack-Nation Global AI Hackathon *August 2025*
Global 24-hour online AI hackathon organized in collaboration with the **MIT Sloan AI Club**, bringing together students and professionals around applied AI challenges.
Contribution: Developed a minimal prototype inspired by *AlphaFold* for protein structure prediction.

Excellence Scholarship from the Graduate Program in Computer Science - PSL Research University *June 2025*
Awarded by the École Normale Supérieure's (ENS-PSL) Master's Committee for outstanding academic performance to admitted students in the year 2025–2026 of the MODO Master specialization (Modeling, Optimization, Decision & Organization).

2025 SpaceHACK for Sustainability *April 2025*
Two-day hackathon co-organized by **Arizona State University** (Library Unit & Interplanetary Initiative), the **University of Toronto** and **CY Cergy Paris University**.
Selected track: "Environmental Migration in Sahelian Africa" — using satellite data to analyze climate dynamics in the Sahel and propose proactive solutions for communities.

Interdisciplinarity Award, Hi!ckathon 2024 *December 2024*
Awarded by Vinci, a global leader in concessions and construction, during the Hi!ckathon, organized by Hi! PARIS, a center of excellence created by:

- **HEC Paris** (Ranked #1 Business School in Europe - Financial Times 2024).
- **Institut Polytechnique de Paris (IP Paris)**, comprising top-tier French engineering schools such as École Polytechnique (#1 Engineering School in France - QS 2025).
- **Inria**, a leading research institute in digital science and AI.

Our project, AquaShield, was selected out of 59 teams and approximately 320 participants, and was recognized for excellence in interdisciplinary collaboration across data science, water resource management, and financial modeling.

Skills

Programming: Python, Java, JavaScript, C, SQL, R, HTML/CSS, Julia
Tools: NumPy, Matplotlib, Spring, MySQL, Postman, JWT, Angular, UNIX/Linux, Microsoft Office
Languages: Arabic (native), French (bilingual), English (fluent), Spanish (intermediate), German (beginner)
Soft Skills: Problem-solving, Communication, Leadership, Teamwork, Time Management